



8.5.1 Additional homework ¶

Homework 8.5.1.1. When using iterative methods, the matrices are typically very sparse. The question then is how to store a sparse matrix and how to perform a matrix-vector multiplication with it. One popular way is known as **compressed row storage** that involves three arrays:

- 1D array `nzA` (nonzero A) which stores the nonzero elements of matrix A . In this array, first all nonzero elements of the first row are stored, then the second row, etc. It has size `nnzeroes` (number of nonzeros).
- 1D array `ir` which is an integer array of size $n + 1$ such that `ir( 1 )` equals the index in array `nzA` where the first element of the first row is stored. `ir( 2 )` then gives the index where the first element of the second row is stored, and so forth. `ir( n+1 )` equals `nnzeroes + 1`. Having this entry is convenient when you implement a matrix-vector multiplication with array `nzA`.
- 1D array `ic` of size `nnzeroes` which holds the column indices of the corresponding elements in array `nzA`.

1. Write a function

```
[ nzA, ir, ic ] = Create_Poisson_problem_nzA( N )  
that creates the matrix  $A$  in this sparse format.
```

2. Write a function

```
y = SparseMvMult( nzA, ir, ic, x )  
that computes  $y = Ax$  with the matrix  $A$  stored in the sparse format.
```